

COORDINATED MOTION PLANNING IS FPT ON DISCRETIZED SIMPLE POLYGONS

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Joint work with Argyris Deligkas (Royal Holloway, University of London), Robert Ganian (TU-Vienna) and Iyad Kanj (School of Computing, DePaul University)

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4. Robots can move simultaneously (in parallel)



Source: <https://www.youtube.com/watch?v=4sEVX4mPuto>

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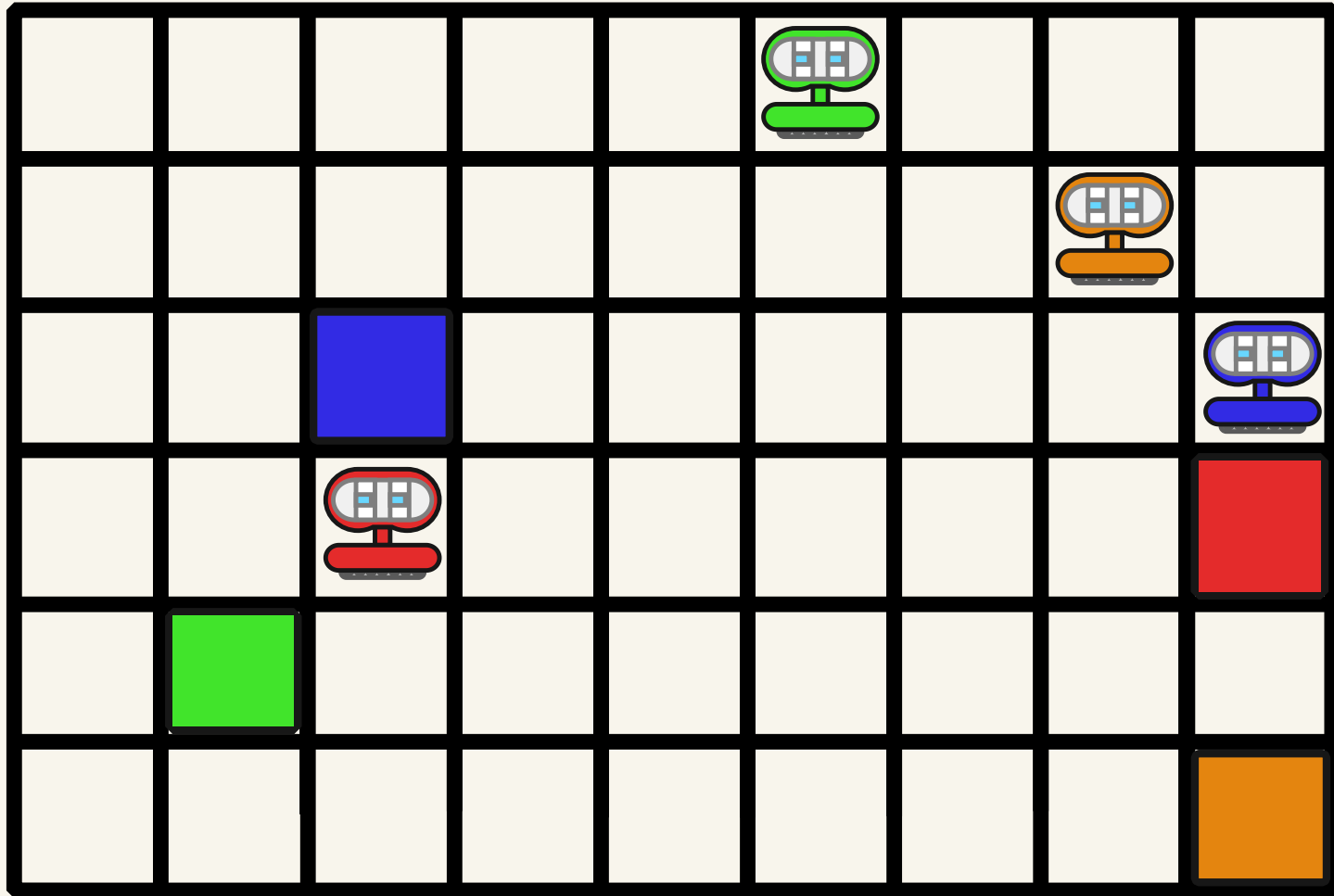
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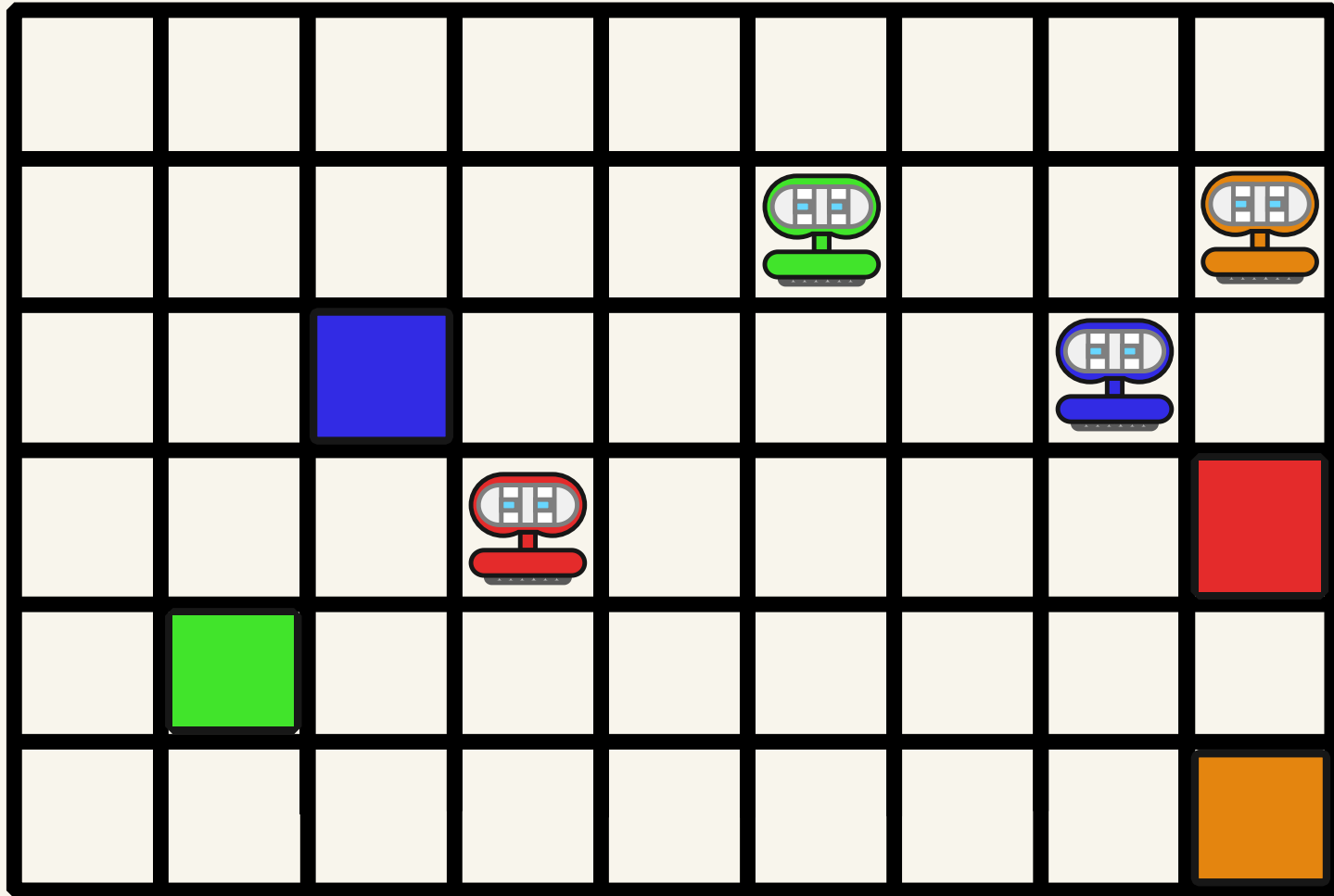
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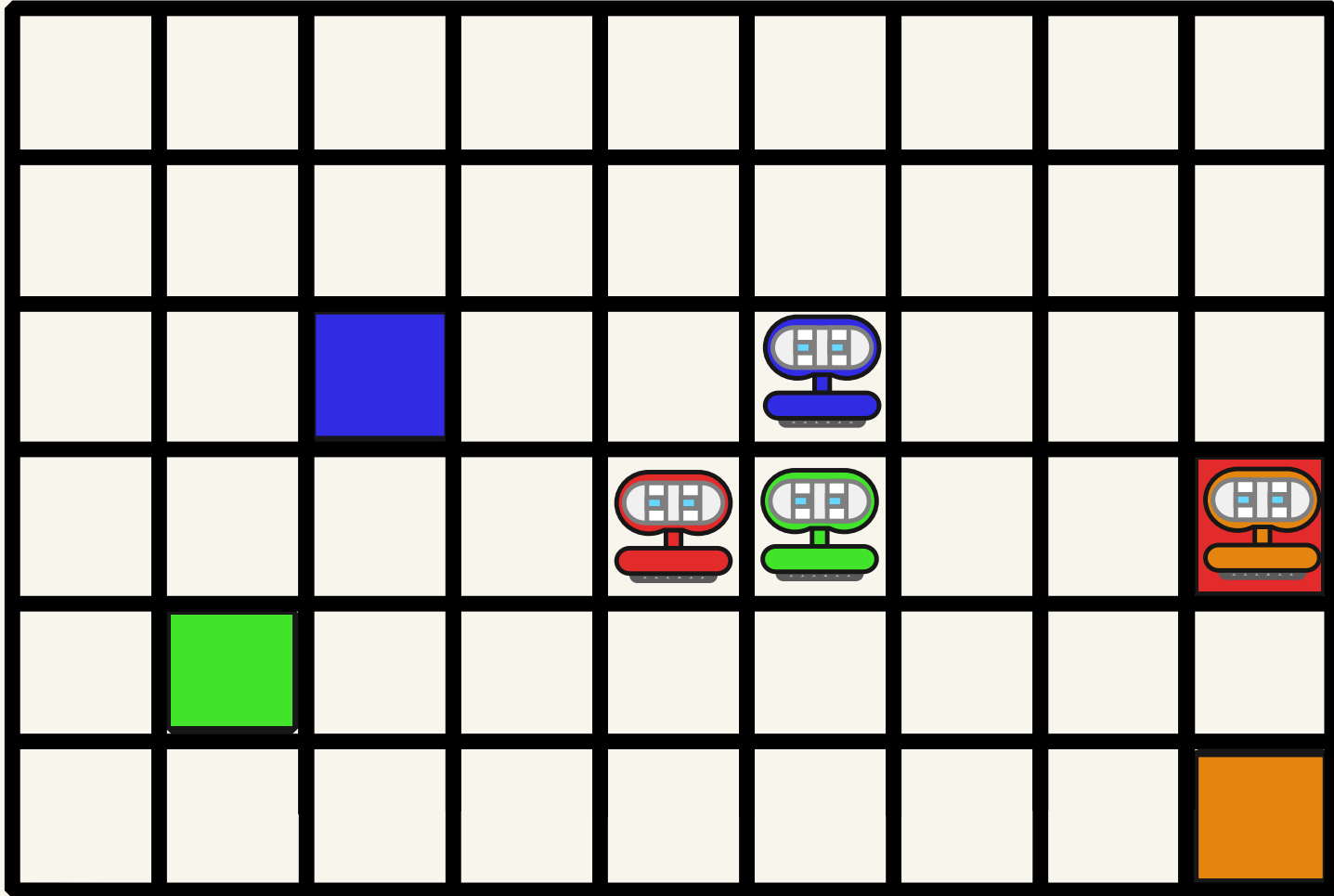
CMP: EXAMPLE



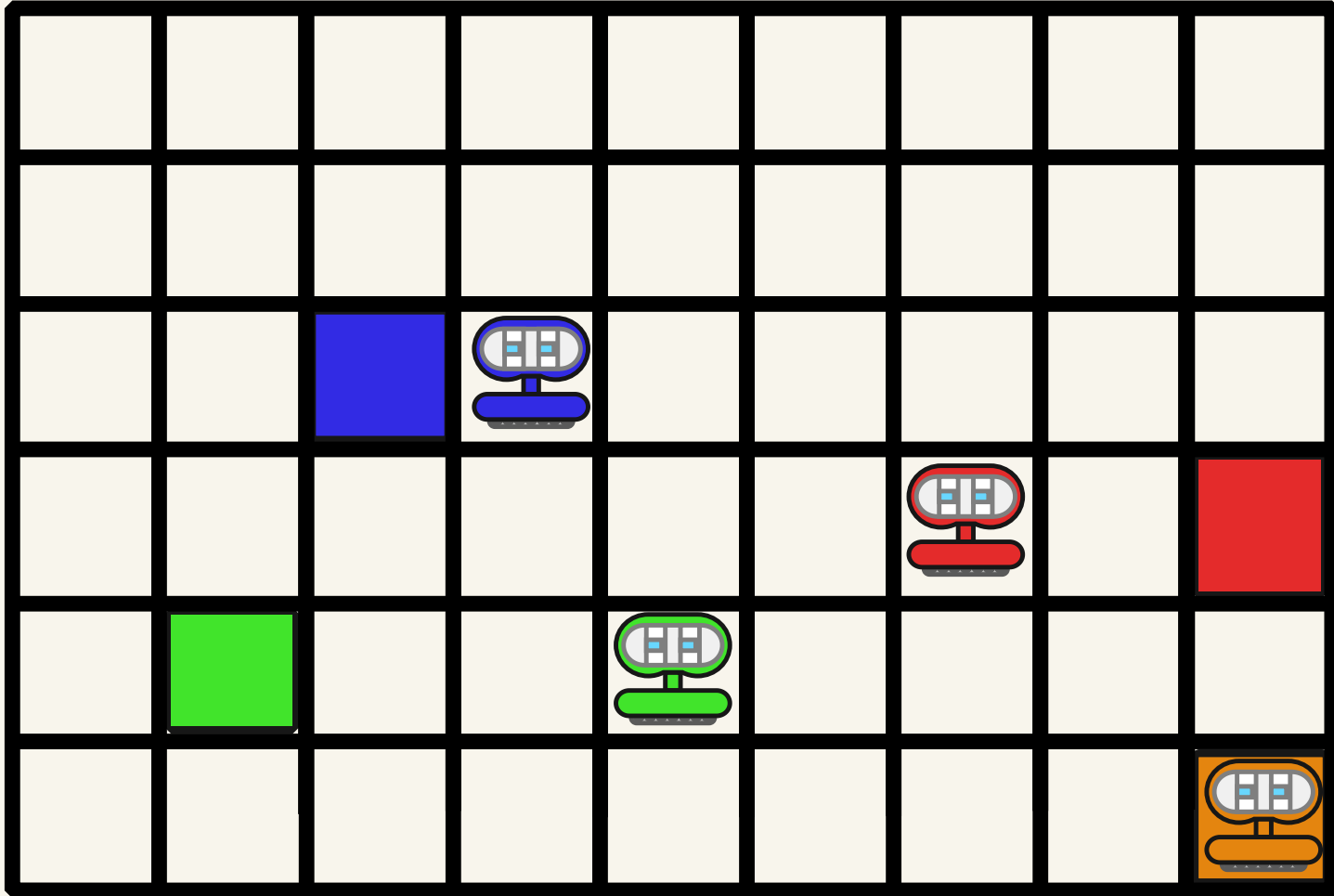
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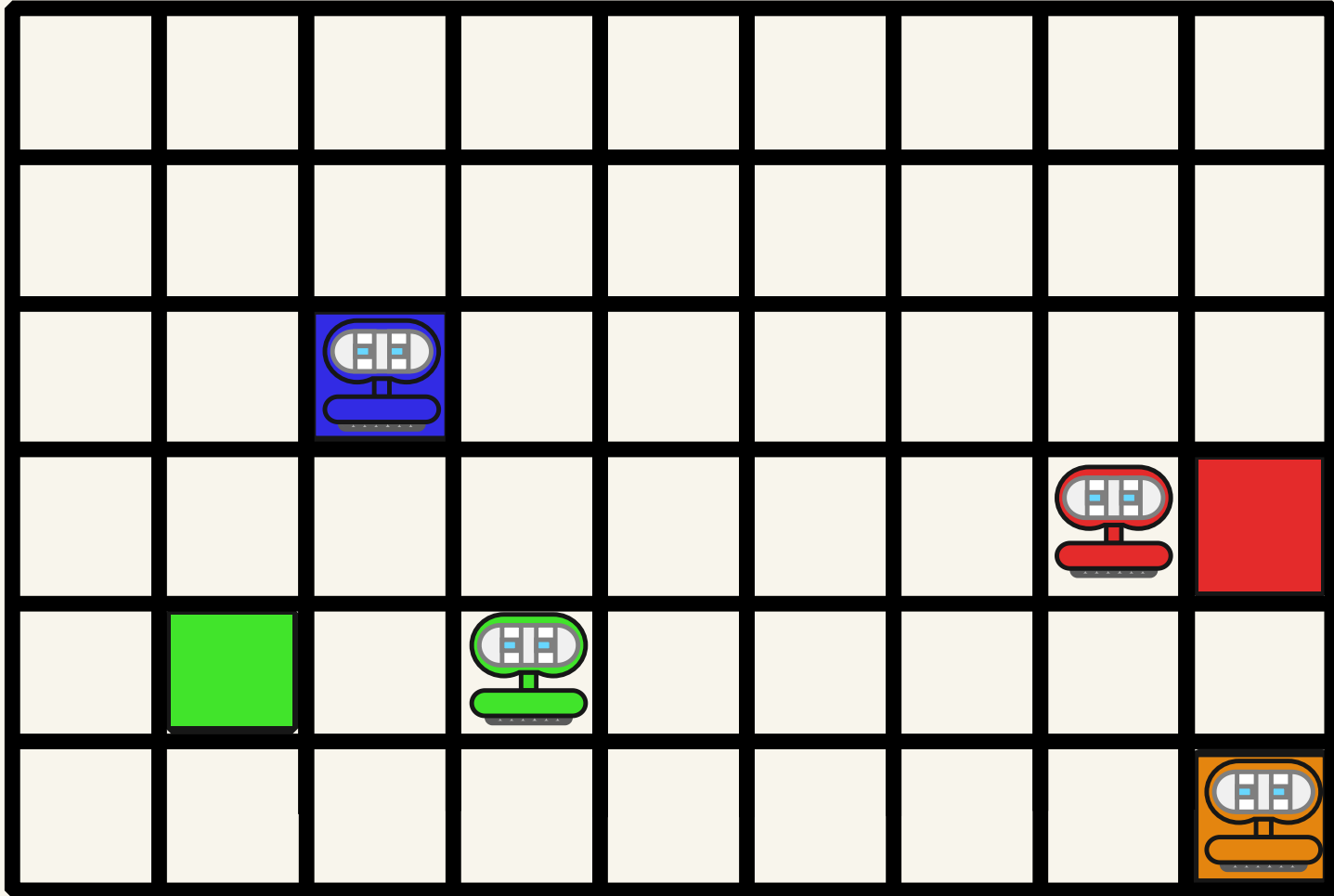
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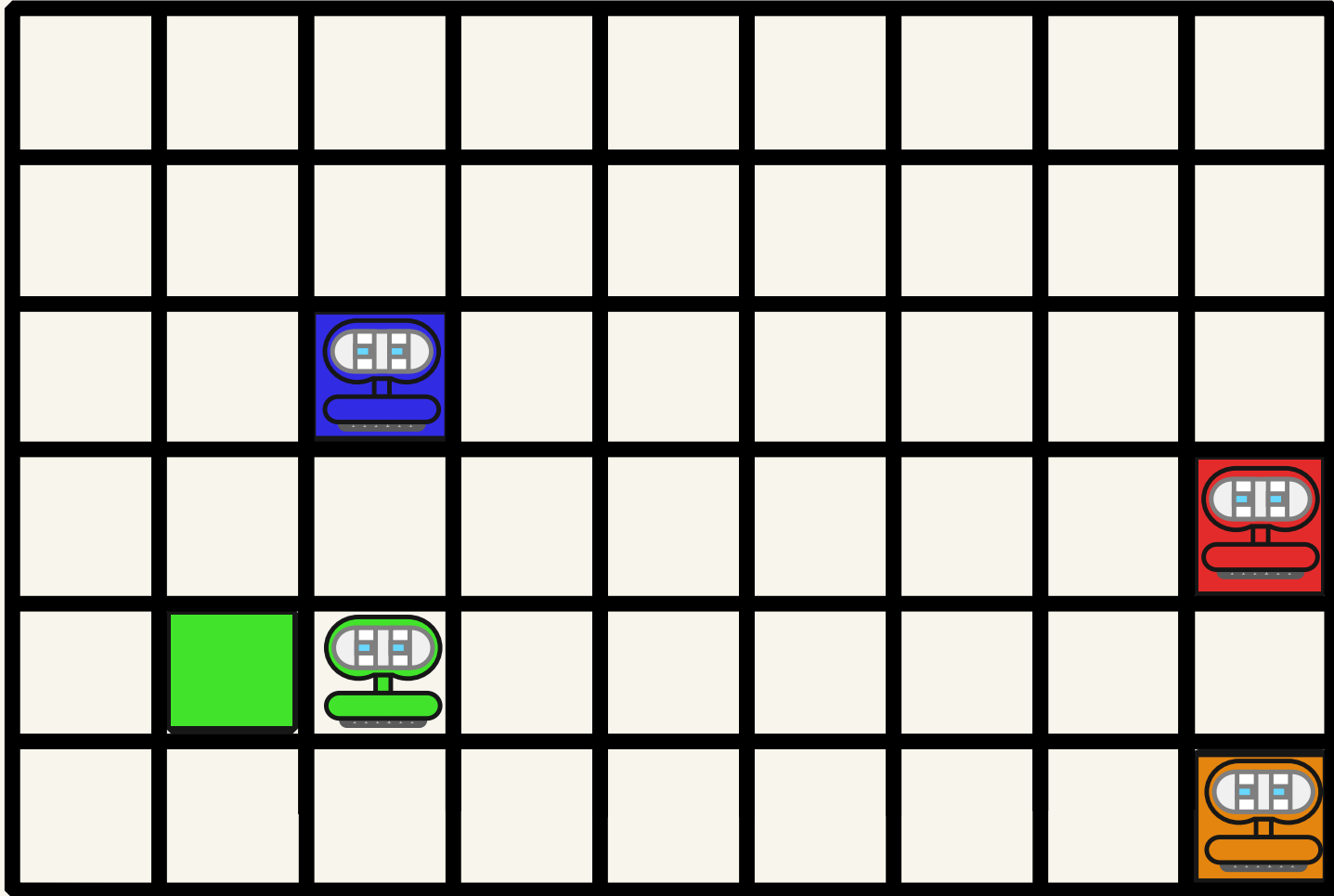
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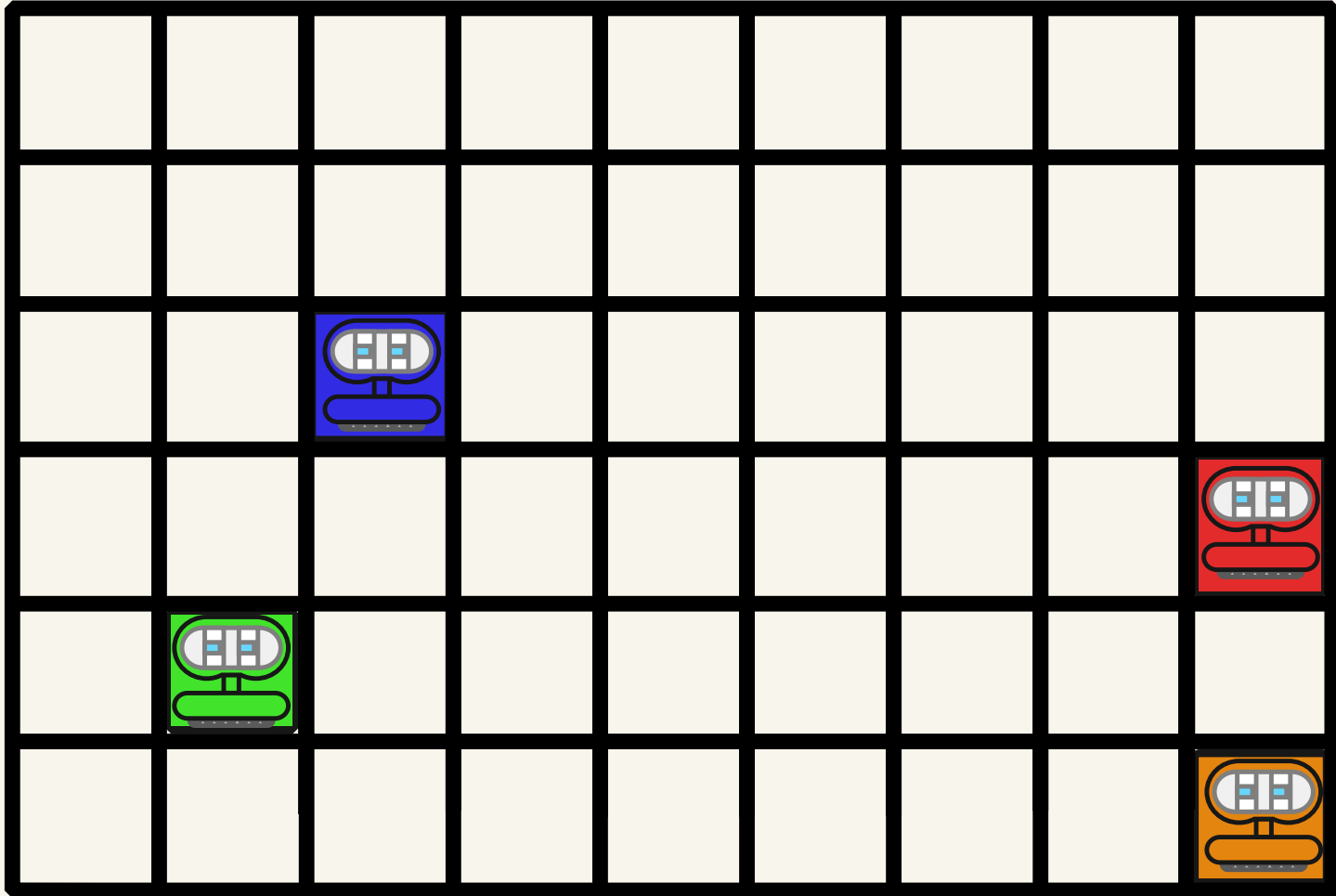
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- Annual workshops affiliated with the top AI conferences (e.g., AAAI, IJCAI) dedicated to CMP
- Posed as The Symposium on Computational Geometry (SoCG 2021) Challenge
- Many algorithms, heuristics, websites, benchmarks, etc, dedicated to CMP

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Most of the settings are NP-hard even on very restricted instances such as grids and trees.

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- *para-NP-hard*: NP-hard for a constant value of the parameter

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- We refer to vertices in $\{s_i, t_i \mid i \in [k]\}$ as *terminals*.

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- The length of a route $W_i = (v_1, \dots, v_\ell)$ is $|\{j \mid v_j \neq v_{j+1}\}|$.
- Two routes *conflict* if the corresponding robots are at *the same vertex* at the end of a time step, or go through *the same edge* (in opposite directions) during the same time step.

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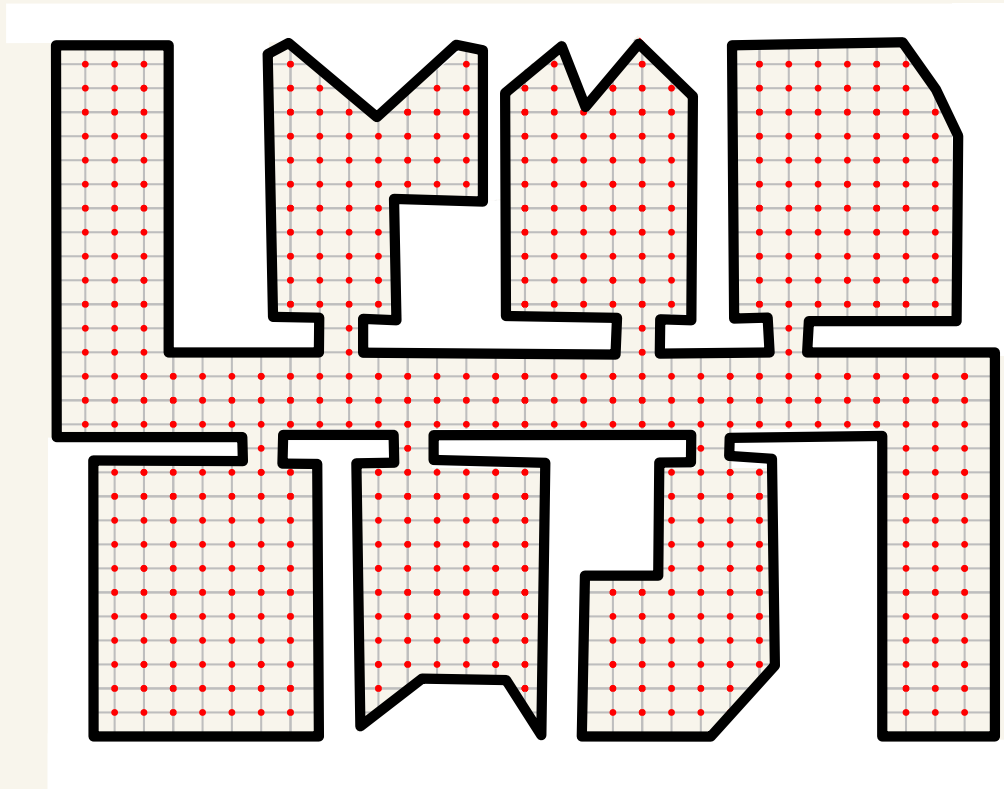
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Question: Is there a schedule for robots of total energy at most ℓ ?

MAIN RESULT

Theorem: CMP is FPT parameterized by k on graphs induced by discretized simple polygons.



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- Decomposition of graph G into sectors.
- A **flattening** of parts of sectors that are “too far”.
- A **weaving** reduction that replaces large grid subgraphs with vertex-disjoint paths.
- Show that resulting graph has bounded **treewidth** and apply known algorithm.

INITIAL PRE-PROCESSING

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Theorem (DEGKR, ICALP'24): An instance of CMP can be reduced in FPT time to an equivalent instance in which $\ell \leq \sum_{i \in [k]} \text{dist}(s_i, t_i) + c \cdot k^5$, for some constant $c > 0$.

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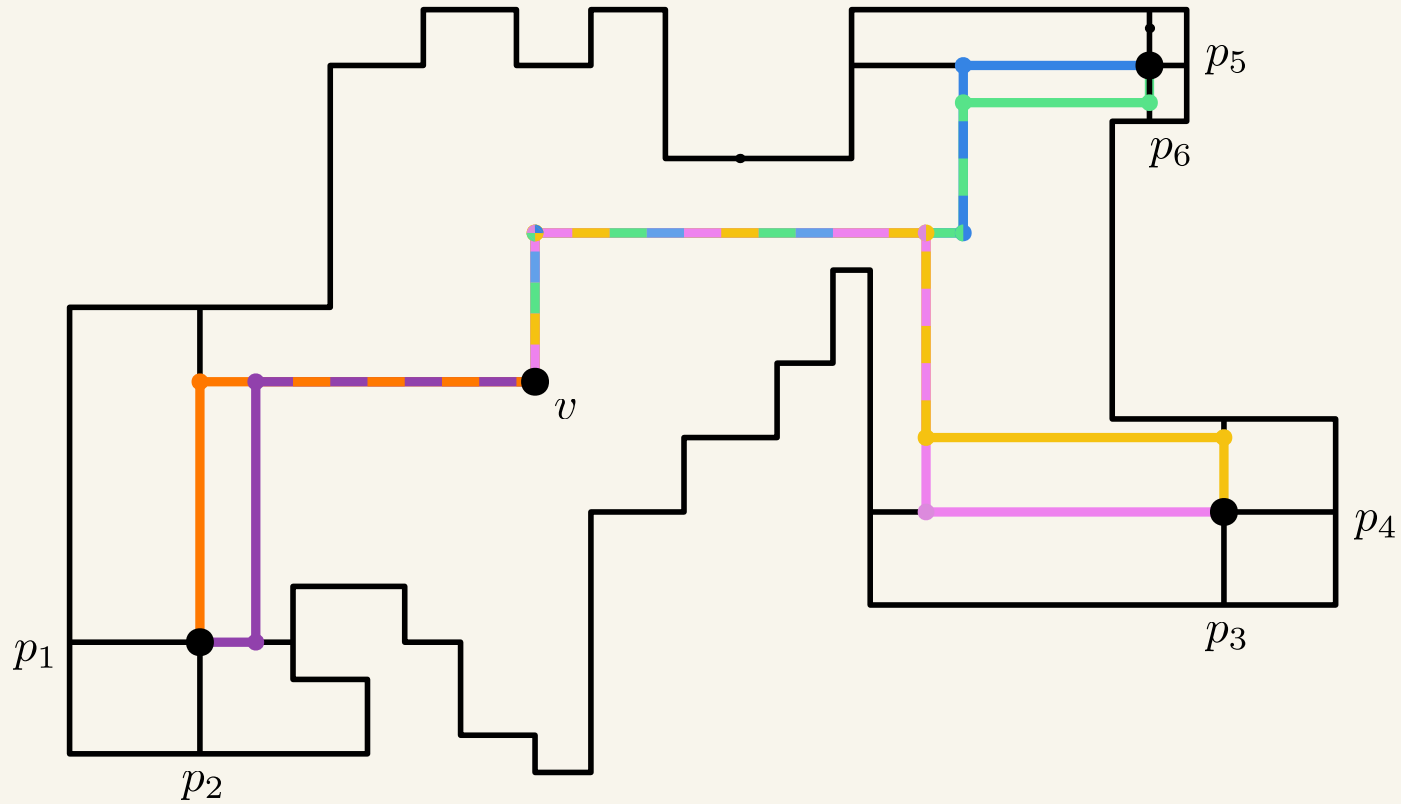
We can assume that :

- No robot R_i has travels more than $\lambda = ck^5$ beyond shortest s_i - t_i path.
- For each robot R_i and for each rectangle (i.e., full rectangular grid) B in G , the number of turns that R_i makes in B plus the number of vertices in B at which it waits is at most $\mu = k^{\mathcal{O}(k^2)}$.

SECTORS AND THE SECTOR GRAPH

Each vertex is associated with a *bend vector*, recording for every terminal and each directional axis d the minimum number of bends on any shortest path from the terminal to that vertex along d .

SECTORS AND THE SECTOR GRAPH



The bend vector of the vertex v is $(2, 1, 4, 3, 3, 4)$.

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- Sectors partition the graph into regions whose vertices have same turn requirements to reach all terminals.
- The sector graph Γ has vertices corresponding to sectors and edges representing adjacencies between them.

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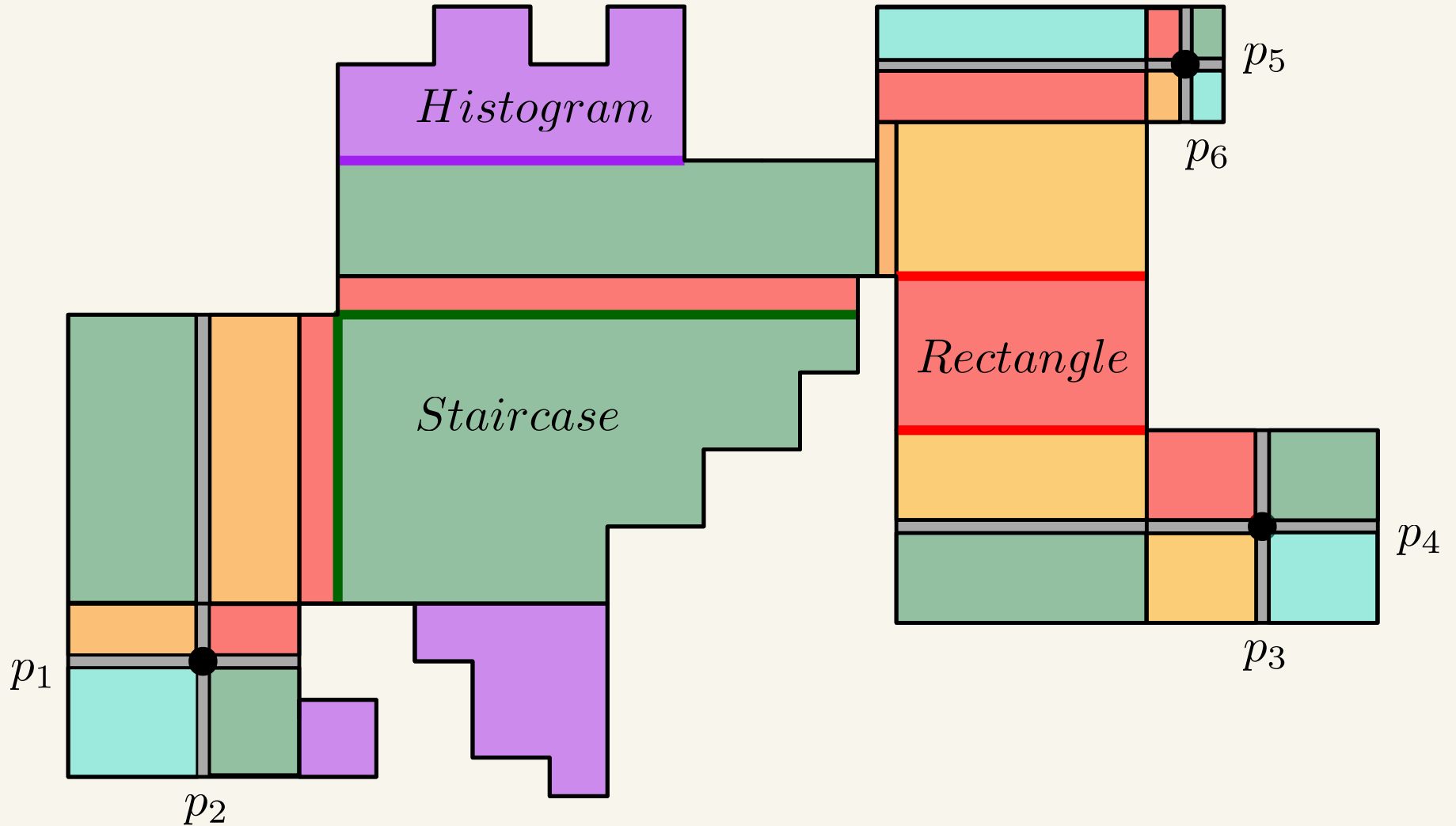
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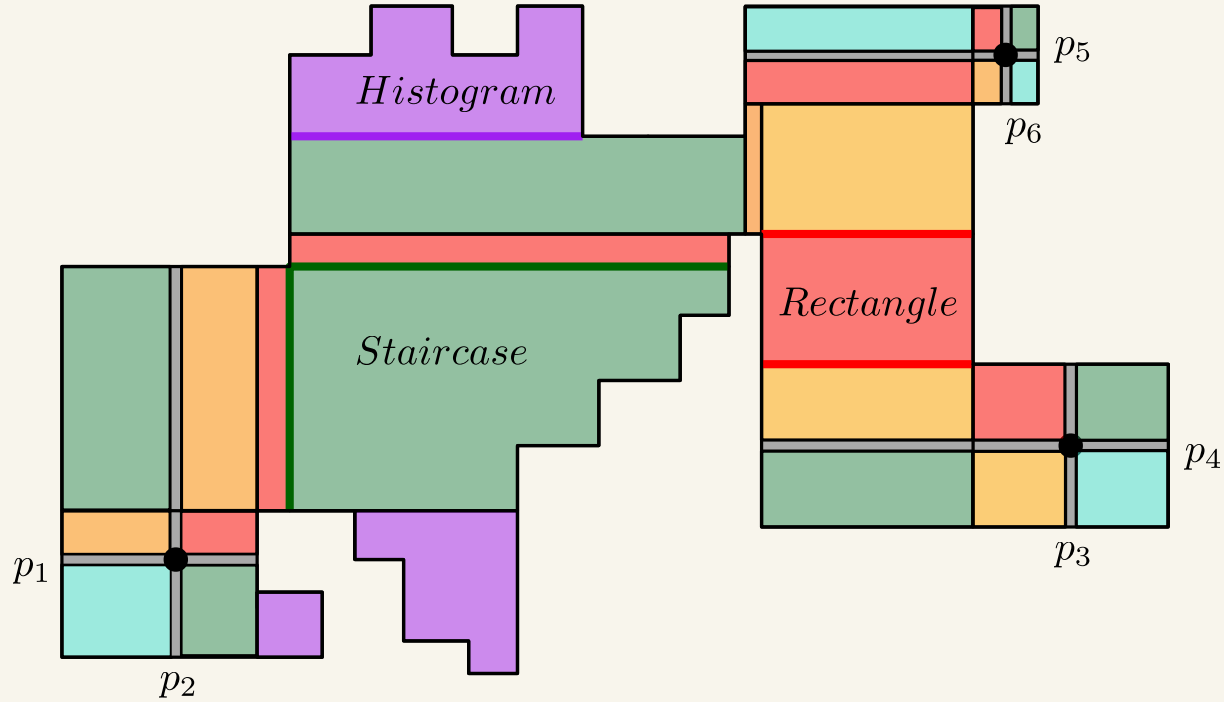
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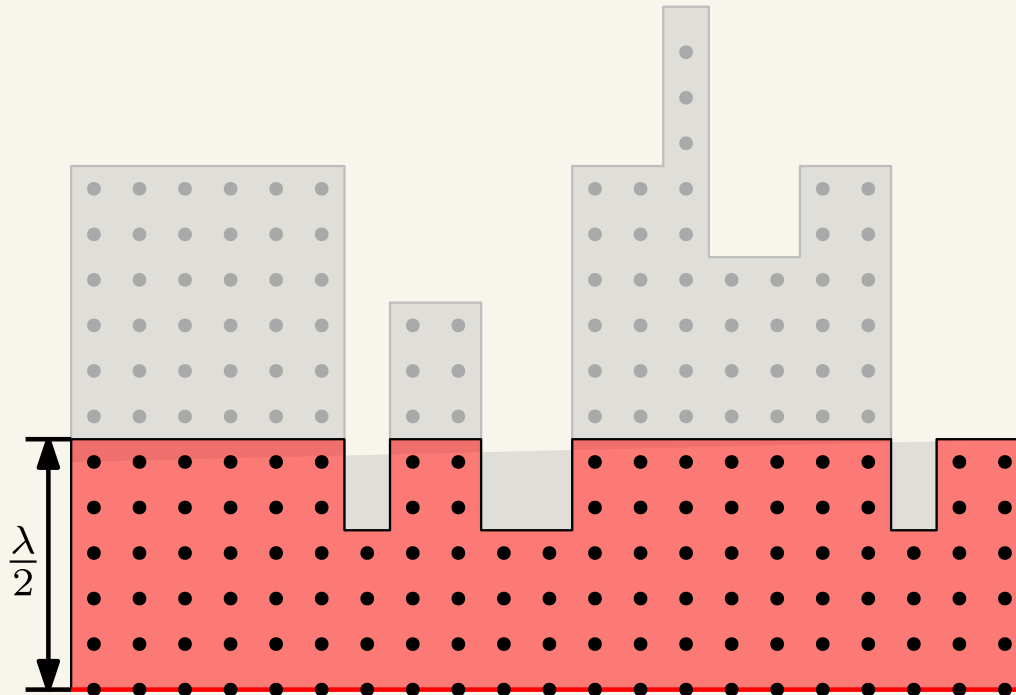


REDUCING HISTOGRAMS

***Lemma:** For any terminal t and a vertex v in a sector S , there is a shortest t - v path that intersects a baseline of S .*

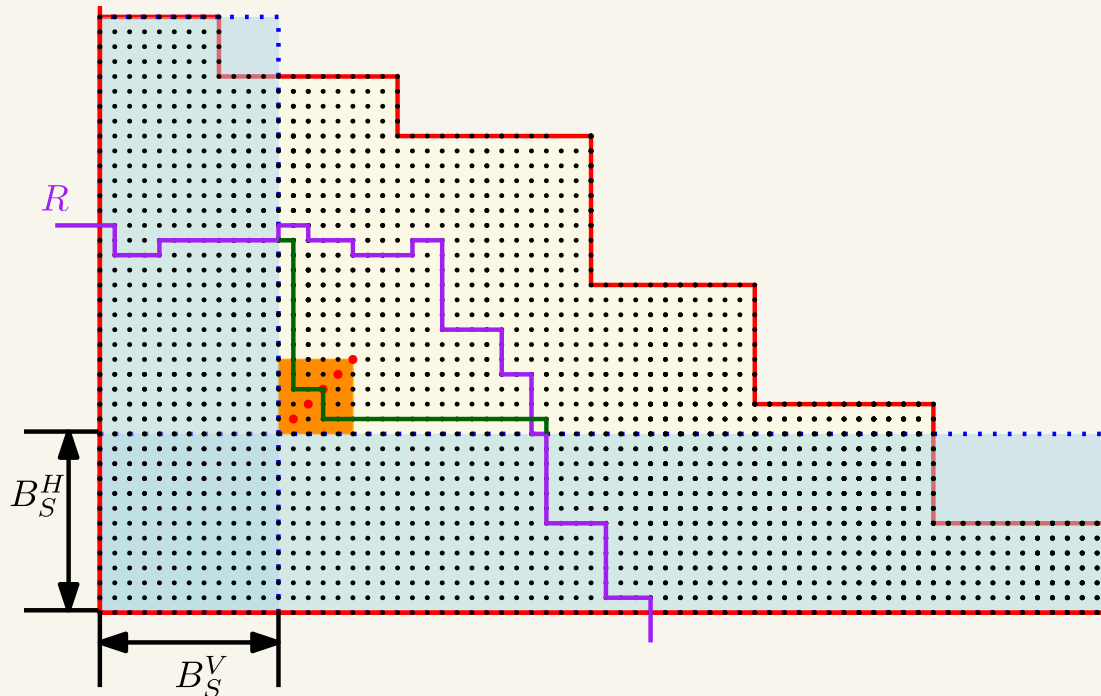
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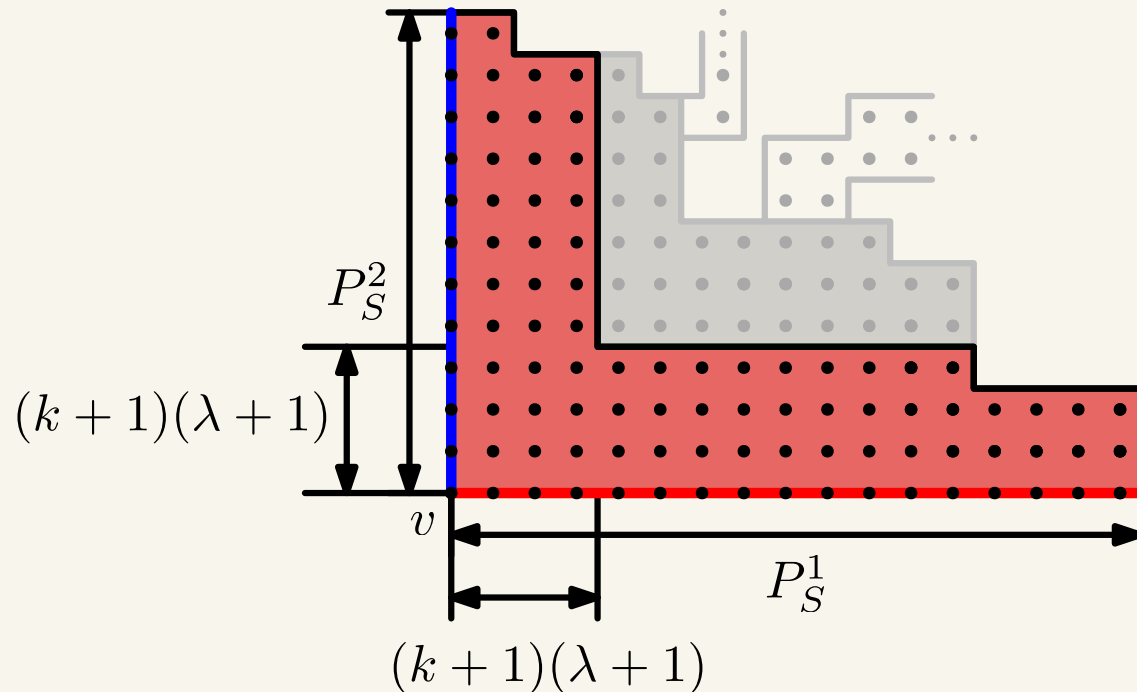
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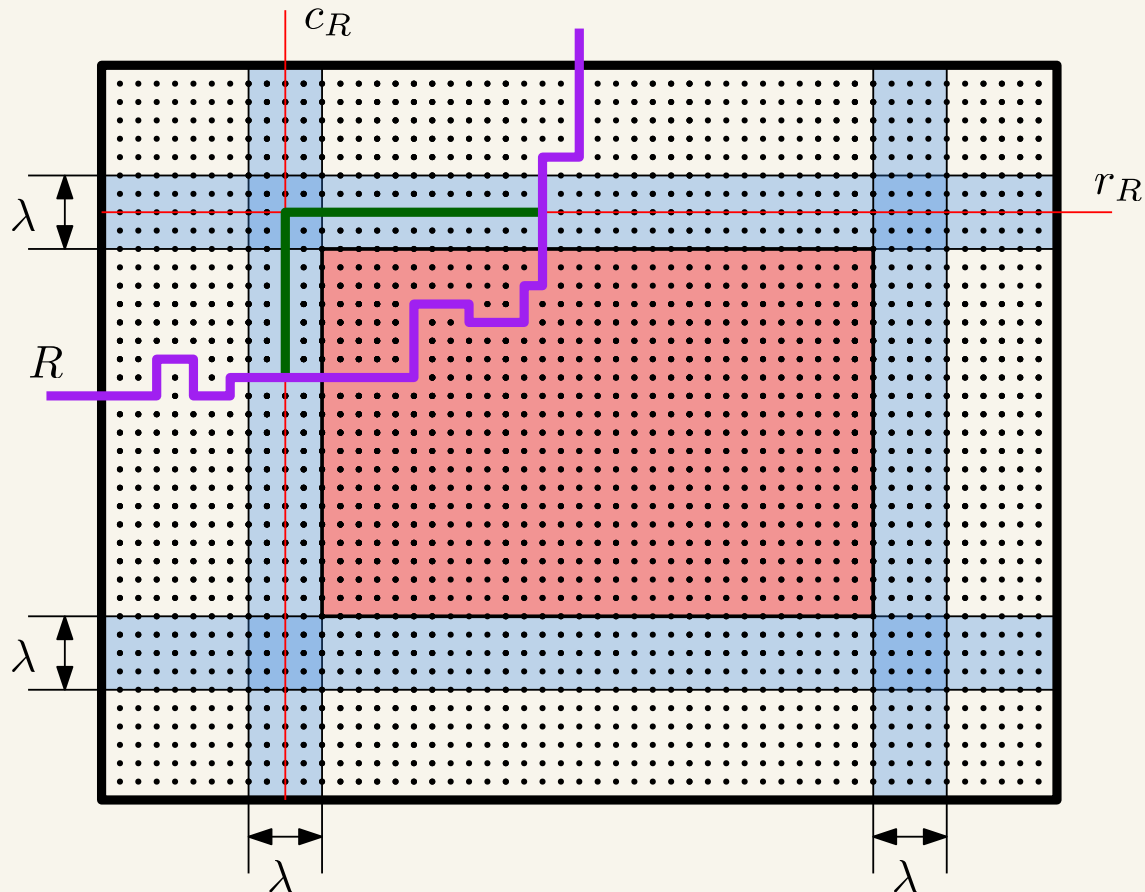
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We prove the lemma in two steps.

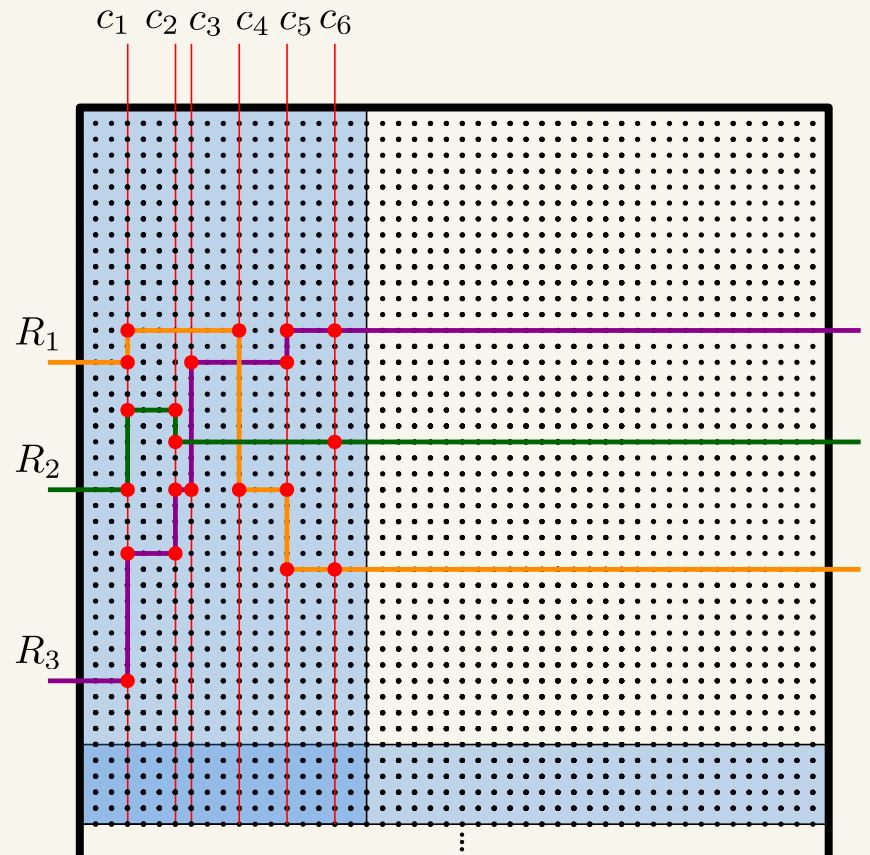
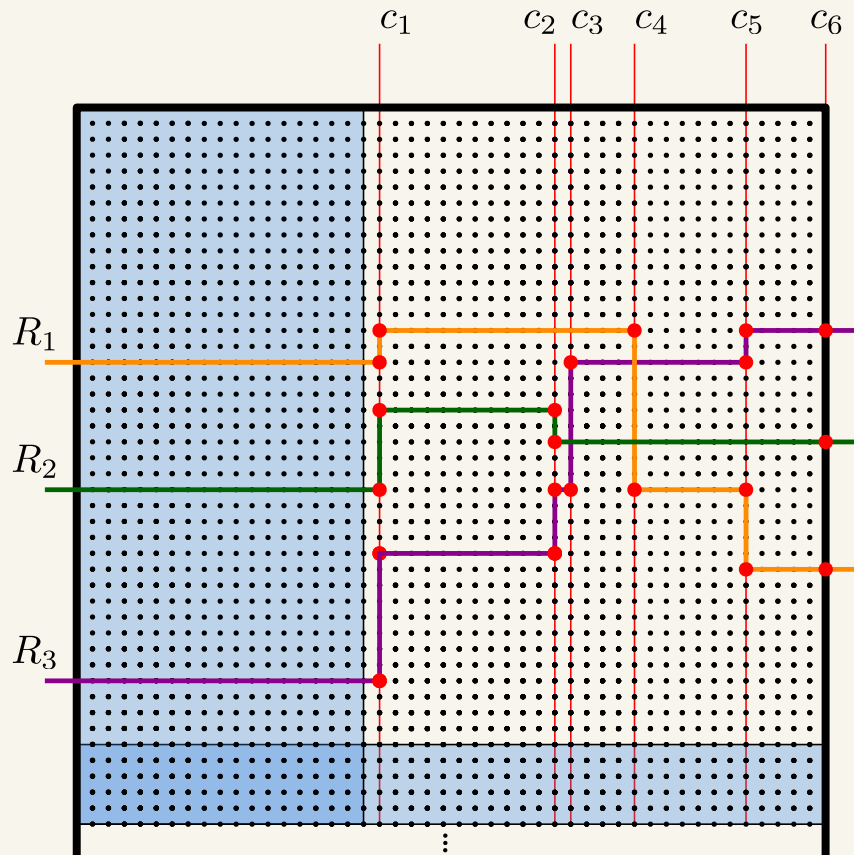
REDUCING RECTANGLE

We first reroute all robots that do not enter and leave in the same direction outside of larger *first frame*.



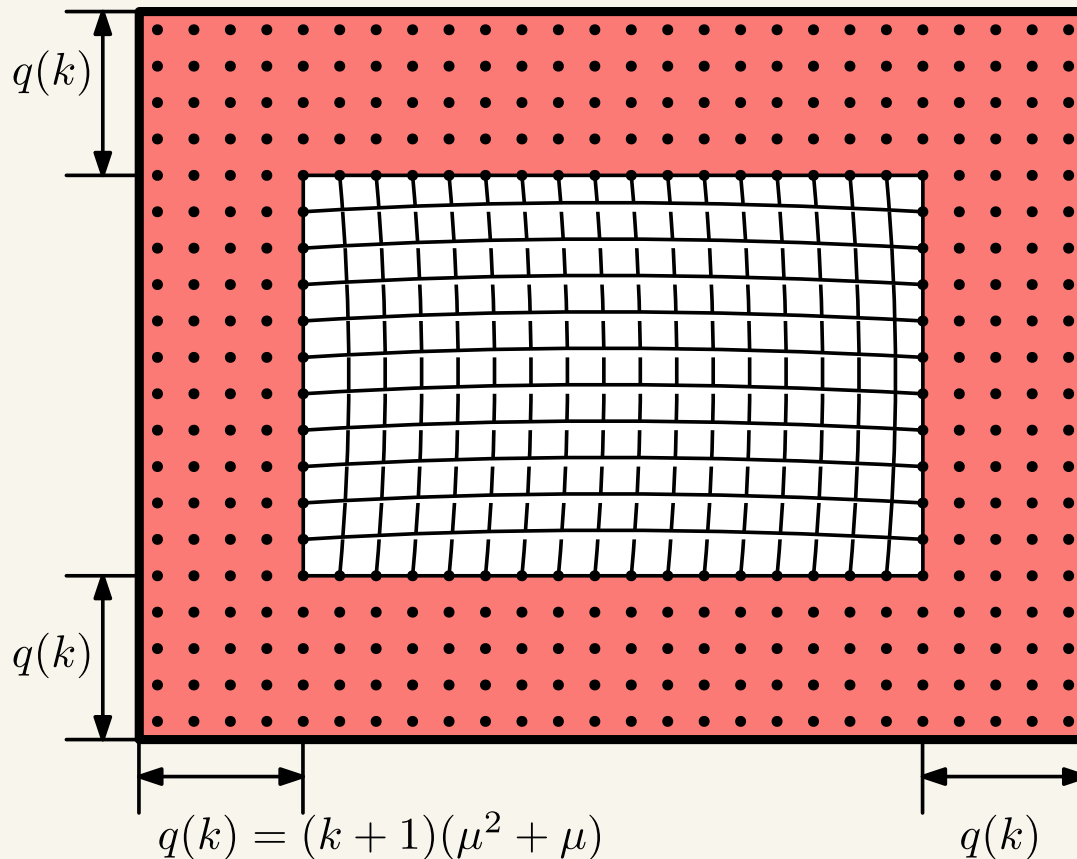
REDUCING RECTANGLE

Then we shift all remaining turns and waiting points inside of second frame to the first frame.



REDUCING RECTANGLE

We replace all internal vertices in the second frame and with degree-2 *weaving paths*.



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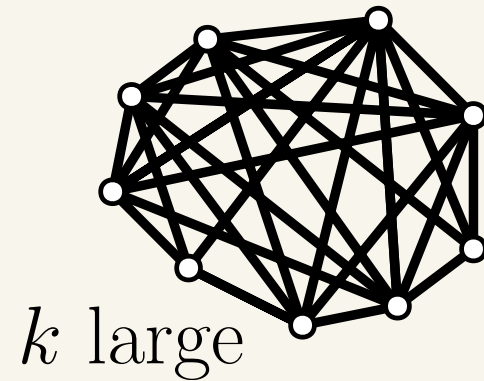
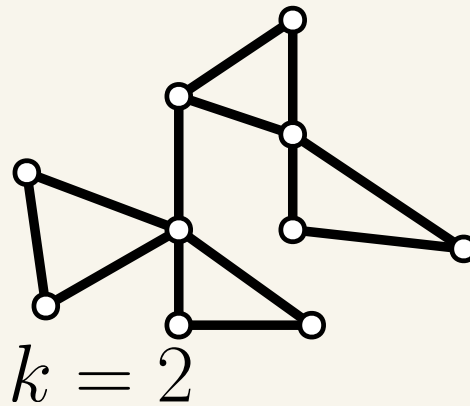
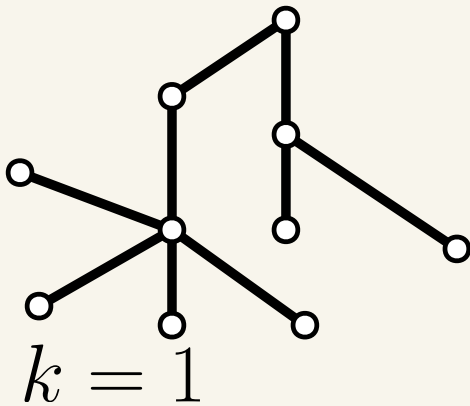
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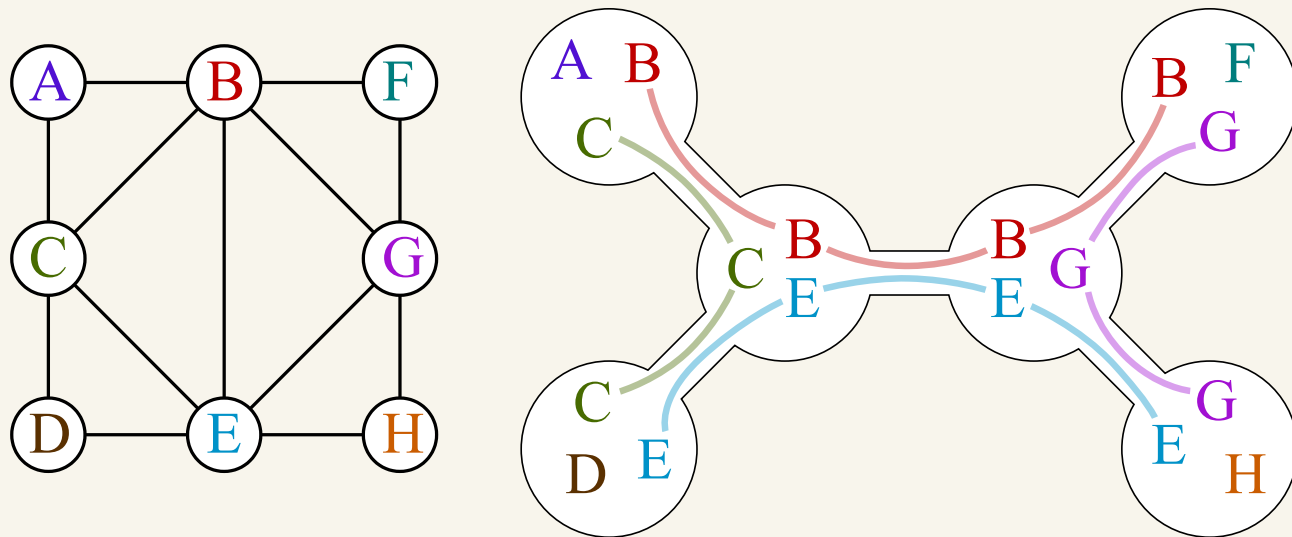
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TREewidth

To determine the treewidth, we decompose the graph into (non-disjoint) parts, called *bags*, that are connected in a tree-like fashion.



By [David Eppstein](#) - Own work, Public Domain, [Link](#)

TREEWIDTH

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Theorem (DEGKR, ICALP'24): CMP is fixed-parameter tractable parameterized by $k + \text{treewidth}$.

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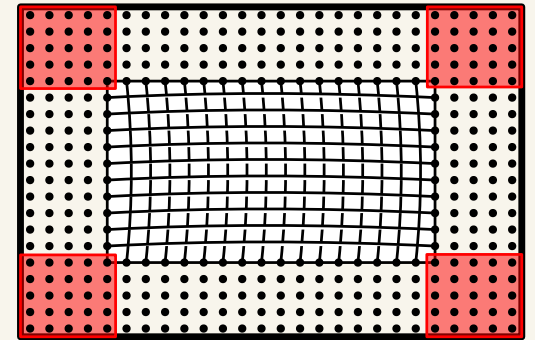
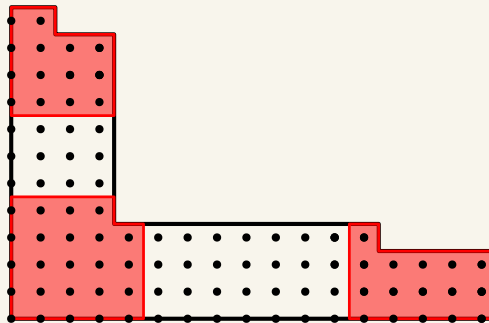
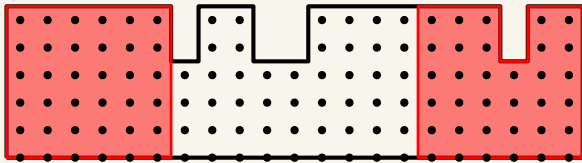
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- Implementation of the FPT algorithms

